set operations

===============
union

union all

intersect

except

union
======

you manage a ecommerce platform
- purchase
- newsletter subscription
- returns

in last 1 month people who interacted in any way

```
select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
UNION
select customer_id, first_name, last_name, email from newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
UNION
select customer_id, first_name, last_name, email from returns
where return_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
```

```
select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
UNION ALL
select customer_id, first_name, last_name, email from newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
UNION ALL
select customer_id, first_name, last_name, email from returns
where return_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
```

UNION
=======

30 RECORDS
UNION
20 RECORDS

ATLEAST 30
MAX 50

UNION ALL
===========
30 RECORDS
UNION ALL
20 RECORDS

50 RECORDS

THE NUMBER OF COLUMNS SHOULD MATCH

AND EVEN THE DATATYPES SHOULD BE COMPATIBLE

// INTERSECT

CUSTOMERS WHO PURCHASED A PRODUCT AND ALSO SUBSCRIBED TO THE NEWSLETTER.

INTESECT

EXCEPT

in mysql both of these were not supported and are supported in latest version.

```
select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
INTERSECT
select customer_id, first_name, last_name, email from newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
```

```
select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
EXCEPT
select customer_id, first_name, last_name, email from newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)
```

=======

3 TABLES - india_employees, us_employees, canada_employees

you want to analyse all this data together

UNION OR UNIONALL?

india_employees
UNION
us_employees
UNION
canada_employees

india_employees
UNION ALL
us_employees
UNION ALL
canada_employees

2 RULES
– NUMBER OF COLUMNS SHOULD BE THE SAME
- THE DATATYPES OF COLUMNS SHOULD BE COMPATIBLE

```
select * from orders;
```

```
select * from customers;
```

find out customers who have not made even a single purchase.

```
select customer_id from customers
except
select order_customer_id from orders;
```

as part of data quality check

I want to find is there any customer who placed a order but not there in customers table.

```
select order_customer_id from orders
except
select customer_id from customers;
```

I want to find out customers who purchased or subscribed to the newsletter but not both in the last 1 month

```
(select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)

except

select customer_id, first_name, last_name, email from
newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH))

union all

(select customer_id, first_name, last_name, email from newsletter_subscriptions
where subscription_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH)

except

select customer_id, first_name, last_name, email from purchases
where purchase_date >= DATE_SUB(current_date(), INTERVAL 1 MONTH));
```

SCRIPTS

```
CREATE TABLE purchases (
   customer_id INT PRIMARY KEY,
   first_name VARCHAR(50),
   last_name VARCHAR(50),
   email VARCHAR(100),
   purchase_date DATE
);


INSERT INTO purchases (customer_id, first_name, last_name, email,
purchase_date) VALUES
(101, 'Aarav', 'Sharma', 'aarav.sharma@example.com', '2024-07-05'),
(102, 'Vihaan', 'Singh', 'vihaan.singh@example.com', '2024-06-25'),
(103, 'Aditi', 'Mehta', 'aditi.mehta@example.com', '2024-07-15'),
(104, 'Rohan', 'Kumar', 'rohan.kumar@example.com', '2024-07-12'),
(105, 'Isha', 'Patel', 'isha.patel@example.com', '2024-07-18'),
(106, 'Kavya', 'Verma', 'kavya.verma@example.com', '2024-06-28'),
(107, 'Arjun', 'Reddy', 'arjun.reddy@example.com', '2024-07-09'),
(108, 'Anaya', 'Nair', 'anaya.nair@example.com', '2024-07-22'),
(109, 'Saanvi', 'Gupta', 'saanvi.gupta@example.com', '2024-07-20'),
(110, 'Kabir', 'Agarwal', 'kabir.agarwal@example.com', '2024-07-19'),
(111, 'Neha', 'Saxena', 'neha.saxena@example.com', '2024-06-24'),
(112, 'Tanishq', 'Rana', 'tanishq.rana@example.com', '2024-07-10'),
(113, 'Mira', 'Bhatt', 'mira.bhatt@example.com', '2024-07-21'),
(114, 'Dev', 'Kapoor', 'dev.kapoor@example.com', '2024-07-17'),
(115, 'Riya', 'Joshi', 'riya.joshi@example.com', '2024-07-11');
```

```sql
CREATE TABLE newsletter_subscriptions (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    subscription_date DATE
);
```

```sql
INSERT INTO newsletter_subscriptions (customer_id, first_name, last_name,
email, subscription_date) VALUES
(101, 'Aarav', 'Sharma', 'aarav.sharma@example.com', '2024-07-01'),
(109, 'Saanvi', 'Gupta', 'saanvi.gupta@example.com', '2024-07-05'),
(116, 'Lakshay', 'Malhotra', 'lakshay.malhotra@example.com', '2024-06-29'),
(112, 'Tanishq', 'Rana', 'tanishq.rana@example.com', '2024-07-03'),
(117, 'Sneha', 'Chawla', 'sneha.chawla@example.com', '2024-07-15'),
(113, 'Mira', 'Bhatt', 'mira.bhatt@example.com', '2024-06-30'),
(118, 'Rahul', 'Pillai', 'rahul.pillai@example.com', '2024-07-06'),
(105, 'Isha', 'Patel', 'isha.patel@example.com', '2024-07-18'),
(106, 'Kavya', 'Verma', 'kavya.verma@example.com', '2024-07-20'),
(115, 'Riya', 'Joshi', 'riya.joshi@example.com', '2024-07-22'),
(119, 'Nikhil', 'Chandra', 'nikhil.chandra@example.com', '2024-07-08'),
(107, 'Arjun', 'Reddy', 'arjun.reddy@example.com', '2024-07-10'),
(108, 'Anaya', 'Nair', 'anaya.nair@example.com', '2024-07-13'),
(114, 'Dev', 'Kapoor', 'dev.kapoor@example.com', '2024-07-14'),
(104, 'Rohan', 'Kumar', 'rohan.kumar@example.com', '2024-07-11');
```

```sql
CREATE TABLE returns (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    return_date DATE
);


INSERT INTO returns (customer_id, first_name, last_name, email, return_date)
VALUES
(103, 'Aditi', 'Mehta', 'aditi.mehta@example.com', '2024-07-17'),
(102, 'Vihaan', 'Singh', 'vihaan.singh@example.com', '2024-07-14'),
(104, 'Rohan', 'Kumar', 'rohan.kumar@example.com', '2024-07-19'),
(110, 'Kabir', 'Agarwal', 'kabir.agarwal@example.com', '2024-07-08'),
(105, 'Isha', 'Patel', 'isha.patel@example.com', '2024-07-21'),
(107, 'Arjun', 'Reddy', 'arjun.reddy@example.com', '2024-07-10'),
(111, 'Neha', 'Saxena', 'neha.saxena@example.com', '2024-07-16'),
(116, 'Lakshay', 'Malhotra', 'lakshay.malhotra@example.com', '2024-07-02'),
(117, 'Sneha', 'Chawla', 'sneha.chawla@example.com', '2024-07-18'),
(112, 'Tanishq', 'Rana', 'tanishq.rana@example.com', '2024-07-07'),
(108, 'Anaya', 'Nair', 'anaya.nair@example.com', '2024-07-09'),
(114, 'Dev', 'Kapoor', 'dev.kapoor@example.com', '2024-07-11'),
(109, 'Saanvi', 'Gupta', 'saanvi.gupta@example.com', '2024-07-12'),
(113, 'Mira', 'Bhatt', 'mira.bhatt@example.com', '2024-07-20'),
(101, 'Aarav', 'Sharma', 'aarav.sharma@example.com', '2024-07-22');
```